

Diary entry for Brown-Representability

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1 Statement of theorem

Definition 1.1 (category of spaces). $\mathcal{C}$ is a **category of spaces** if it consists of pointed CW complexes (or rather, spaces admitting a CW structure) along with all basepoint-preserving continuous maps as the morphisms, as well as having the property that if $X \in \mathcal{C}$ and $X' \subseteq X$ is a subcomplex of X under some CW structure on X , then $X' \in \mathcal{C}$.

Two common categories of spaces we will use: $\mathcal{C}_1$ is the category of all pointed CW complexes and $\mathcal{C}_0$ is the category of all pointed finite CW complexes.

Definition 1.2 (proper triad). A **proper triad** of spaces in a category of spaces $\mathcal{C}$ is a triple $(X_1 \cup X_2, X_1, X_2)$ such that $X_1, X_2, X_1 \cup X_2, X_1 \cap X_2 \in \mathcal{C}$ and such that the pairs $(X_1, X_1 \cap X_2)$ and $(X_2, X_1 \cap X_2)$ have the homotopy extension property.

For $X, Y \in \mathcal{C}$ define $[X, Y]$ to be the set of homotopy classes of pointed maps $X \rightarrow Y$. Then, we have a contravariant functor $[-, Y]_{\mathcal{C}}$ from $\mathcal{C}$ to sets sending $f: X \rightarrow X'$ to $f^*: [X', Y] \rightarrow [X, Y]$ (we will generally omit the subscript $\mathcal{C}$ unless the ambient category isn't clear) via $f^*[g] = [g \circ f]$. A contravariant functor $\mathcal{C} \rightarrow \mathbf{Set}$ that is naturally isomorphic to some $[-, Y]$ is called **representable**, which is why Brown's representability theorem is called as such.

Let $H: \mathcal{C} \rightarrow \mathbf{Set}$ be a contravariant functor. For brevity define the following statements:

- (h) If $f, g: X \rightarrow Y$ are homotopic morphisms in $\mathcal{C}$, then $H(f) = H(g)$.
- (e1) If p is a point then $H(p)$ consists of one element.
- (e2) Let (X, X_1, X_2) be a proper triad. If $u_1 \in H(X_1)$ and $u_2 \in H(X_2)$ restrict to the same element of $H(X_1 \cap X_2)$ (i.e. pull back via inclusions to the same thing) then there is an element $v \in H(X)$ which restricts to u_1 and u_2 . Furthermore, if $X_1 \cap X_2$ consists of one point, then v is unique.
- (c) $H(S^n)$ is countable for all $n > 0$.
- (w) Let $\bigvee_{\alpha} S_{\alpha}^n$ be the wedge of a collection of spheres and $i_{\beta}: S_{\beta}^n \rightarrow \bigvee_{\alpha} S_{\alpha}^n$ the inclusion. Then,

$$\prod_{\alpha} H(i_{\alpha}): H\left(\bigvee_{\alpha} S_{\alpha}^n\right) \rightarrow \prod_{\alpha} H(S_{\alpha}^n)$$

is an isomorphism.

- (l) (I don't get this one) Suppose $X_1, X_2, \dots$ is a collection of subcomplexes of $X = \bigcup_{n=1}^{\infty} X_n \in \mathcal{C}$ with respect to some CW structure on X such that $X^n = X_n$ (i think? paper says X_n^n). Then there are inclusions $X_n \rightarrow X_m$ for $m \geq n$ which make the X_n into a directed system, and

$$\varprojlim H(i_n): H(X) \rightarrow \varprojlim H(X_n)$$

is an epimorphism. (I think limit makes sense? The X_i colimit to X and then H flippy flips it)

Lemma 1.3. If $\mathcal{C} = \mathcal{C}_0$ and H satisfies (c), then H satisfies (w) and (l).

Theorem 1.4 (Brown's representability theorem). If $H: \mathcal{C} \rightarrow \mathbf{Set}$ is a contravariant functor and either

- (a) $\mathcal{C} = \mathcal{C}_0$ and H satisfies (h), (e), and (c); or
- (b) $\mathcal{C} = \mathcal{C}_1$ and H satisfies (h), (e), (w), and (l),

then there is a space $Y \in \mathcal{C}_1$ which is unique up to homotopy type such that $[-, Y]$ is naturally isomorphic to H .

How to construct such a Y ?